

The Global Perspective: AI-driven Fraud Trends, Threats, and Insights

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The democratization of AI enables faster, cheaper, and more scalable attacks, while disrupting the traditional customer behavior patterns used to detect fraud



1 in 4

Organizations in 2024 that **experienced deepfake fraud** incidents¹



\$40bn

in estimated **fraud losses enabled by GenAI** in the US alone by 2027²



3%

E-commerce and retail **merchants** feel **well prepared to manage AI-enabled fraud**³

Prominent AI-Enabled Threats



Deepfake toolkits / Fraud-as-a-Service / Refunds-as-a-Service

- **End-to-end fraud offerings** are becoming more commonplace on the dark-web marketplace, including sale of valuable source data, synthetic identities with account history and **Image injection & emulator tools** (to spoof liveness detection & mimic legitimate device activity).
- **Refund fraud as a service** via fake image/video of damaged products. Tor network usage surged past 3 million daily users in 2025⁴.



Mass-personalized social engineering & Authorized Push Payment (APP) Fraud

- **Use of AI to add legitimacy to scams** (e.g., investment scam website, company history, activity etc.) is **boosting APP fraud**, estimated to surge to nearly **\$15 billion by 2028**⁵ from estimated \$8.3 billion in 2024.
- Autonomous agents are also **facilitating highly personalized phishing attempts**, scraping web for personal data and an understanding of the psyche of their target.



Opportunistic exploitation of Agentic Commerce for fraud and money laundering

- The rise of Agentic Commerce, in a largely unfettered ecosystem is driving an uptick in **fake / manipulated online storefronts to dupe agents**, as well as signal spoofing (click fraud and manipulated reviews).
- **Concerns rising over agent-to-agent** micro-transactions to layer proceeds of crime via refunds / returns etc.



Insider risk from both orchestrated attacks, and unintentional AI misuse

- Increased efforts to **embed malicious employees** within organizations through synthetic credentials, spoofed callbacks/interviews.
- Heightened risk of insiders manipulating AI/agents e.g., with advanced prompt injection or **Denial-of-Wallet attacks**.
- Increased security risk posed by **unintentional misuse of AI** (e.g., risks of **AI-generated source code** introducing backdoors into controls deployed by Fraud teams).

¹ 2024 Deloitte Survey on GenAI - [generative-ai-and-the-fight-for-trust.pdf](#)

² 2025 Deloitte Article on Emerging GenAI Risks - [Managing gen AI risks | Deloitte Insights](#)

³ 2025 Deloitte Survey on Agentic AI Fraud in Retail

⁴ 2025 Tor Article on Dark Web Trends and Usage - [Dark Web Statistics 2025: Inside the \\$470M Underground](#)

⁵ 2025 Deloitte Article on APP fraud - [The rise of authorized push payment fraud | Deloitte Insights](#)

Note: In an aggressive scenario, where AI-driven APP fraud incidents outpace societal and institutional defenses, APP fraud losses could shoot up to as much as **\$18.2 billion by 2028**

Emerging Trends on Agentic Payment and the Fraud Risk Implications

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Agentic AI will make retail fraud smarter and faster – fraudsters will use AI agents that learn and adapt in real time⁶

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ESTABLISHED NORTH AMERICAN RETAILER

IN-SESSION POLL RESULTS!

What are your organization's **primary concerns** around agentic payments / commerce?

39%

Fraud detection and dispute management

27%

Misuse of customer data and increased attack surfaces due to AI agents / workflows

14%

Obfuscated transaction data and limitations to differentiate good vs. bad bots

10%

Fragmented protocols & hampered traceability of actions and reasoning

10%

Agent authorization / consent to perform payment on behalf of consumers

0%

Other

As Agentic AI transforms eCommerce, AI agents can independently search products, compare prices and initiate payment through customer delegation. With over 58% consumers likely to use AI for gift shopping this year⁷, an innovative approach must be taken by risk management to enable this new consumer behavior.

Key Focus Areas for Banks

Multiple Agentic Payment Protocols & Frameworks

- **Multiple frameworks and protocols** being developed (e.g., OpenAI, Google, Mastercard, Visa) instead of one universal standard
- **Ongoing rapid changes** coupled with an anticipated surge in holiday season transaction volume

- Each agent platform may require **bespoke risk models** and contractual and privacy terms
- Inconsistent data sharing protocols
- Potential for **inconsistent dispute handling** and higher operational costs

Data Obfuscation and Behaviors Shift

- Obfuscated **digital telemetry** (session length, etc.) on agent-initiated payment can reduce the efficiency of the existing fraud detection engine
- Change of user and entity behavior may lead to **falsely associating legitimate interaction** with bot attack

- **A need to delineate fraud monitoring strategy** between agent-based transactions from the existing human-based transactions
- **Lose visibility** into key behavioral and transactional data traditionally used for fraud scoring (e.g., device fingerprint, IP address)
- **Increased attack surface** introduced by third parties (e.g., agent providers, fraud solution providers)

Liability & Consent Management

- **Evolving consent and liability framework** through RPAA, Bill C-27 (proposed AI and Data Act)
- **Disputes** from agent error, malicious agents, compromised agents via prompt injection

- **Ambiguity** in liability allocation for unauthorized or erroneous agent-initiated transactions
- A need for a stronger authorization standard including tokenization for payment security
- **Complex dispute** resolution processes when agent actions are opaque

⁷ 2025 Riskified Article on Agentic Commerce Risks - [Global Study: 73% of Shoppers Using AI in Shopping Journey - But Merchants Face New Agentic Commerce Risks](#)

Banks must evaluate key considerations and formulate responses to Agentic Commerce, including a fraud management strategy that effectively addresses their overall needs



Standardized consent capture / incorporate tokenization authorization

Require agent platforms to provide verifiable proof of consumer intent (e.g., digital signature, voice consent logs). Consumers must define transaction parameters (e.g., spending limits, merchant categories) before enabling agent transactions. Utilize datapoints generated in the payments flow to trace authentication of agentic transactions



Enhance transaction data and use of external data intelligence

- Enhance transaction data with agent risk signals & identification; leverage cryptographically secure tokens that uniquely identify AI agents during transactions. Integrate tokens into transaction metadata to flag agent-driven purchases
- Use of consortium information from eCommerce ecosystem / merchants to minimize false declines



Enhanced fraud monitoring & detection

- Delineate detection strategy between human and AI based transactions
- Build agent-specific anomaly detection models, thresholds and feedback loops to identify good bot (agentic) and bad bot (traditional threat vector)



Augment dispute management and customer notification

Explore real-time notifications, or workflows allowing customers to review and override agentic actions; develop audit trails, override policies and maintain logic transparency to ensure compliance and fairness in Agentic Commerce transactions; balancing customer experience, governance and disputes exposure

Global Response to AI-Powered Capabilities For Fraud Risk Management

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Ultimately, implementing AI wasn't just a technical shift – it required balancing innovation with regulatory rigor, operational resilience and strong internal collaboration⁸

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SVP of TECHNOLOGY & REPORTING, CITI BANK

IN-SESSION POLL RESULTS!

What are the **key blockers** to your roll out and adoption of GenAI for fraud management?

20%

Budget constraints and competing priorities

17%

Poor data quality / lineage / orchestration

7%

Resource capacity and skills

41%

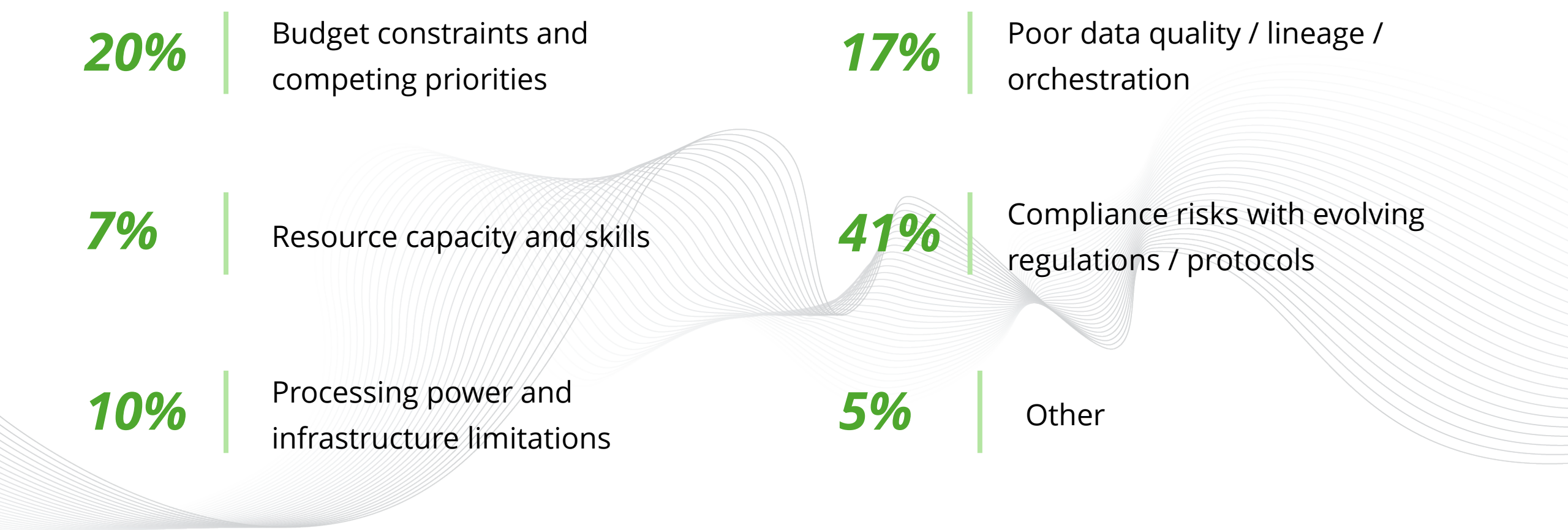
Compliance risks with evolving regulations / protocols

10%

Processing power and infrastructure limitations

5%

Other



Across key regions, organizations are looking to strategically invest in AI-powered capabilities to augment capabilities in a holistic and practical manner

Capability Play: Canada

- Leading institutions are exploring the use of AI tools to train and boost **insider threat prevention** capabilities (e.g., including insider risk within the corporate security teams themselves) via:
 - **Red teaming exercises to exploit at-risk applications**, identifying weak controls
 - Improving **User and Entity Behavioral Analytics (UEBA)** in order to establish a 'normal' baseline for each employee and monitor access pattern behavior for anomalies against baseline and peers
 - **Advanced sentiment analysis** in email correspondence

Capability Play: UK

- Large banks leveraging enhanced AI detection capabilities for **document reviews** (to identify fraudulent/synthetic credentials) as well as **deepfake detection** in the relationship manager channel

Customer Experience Play: UK

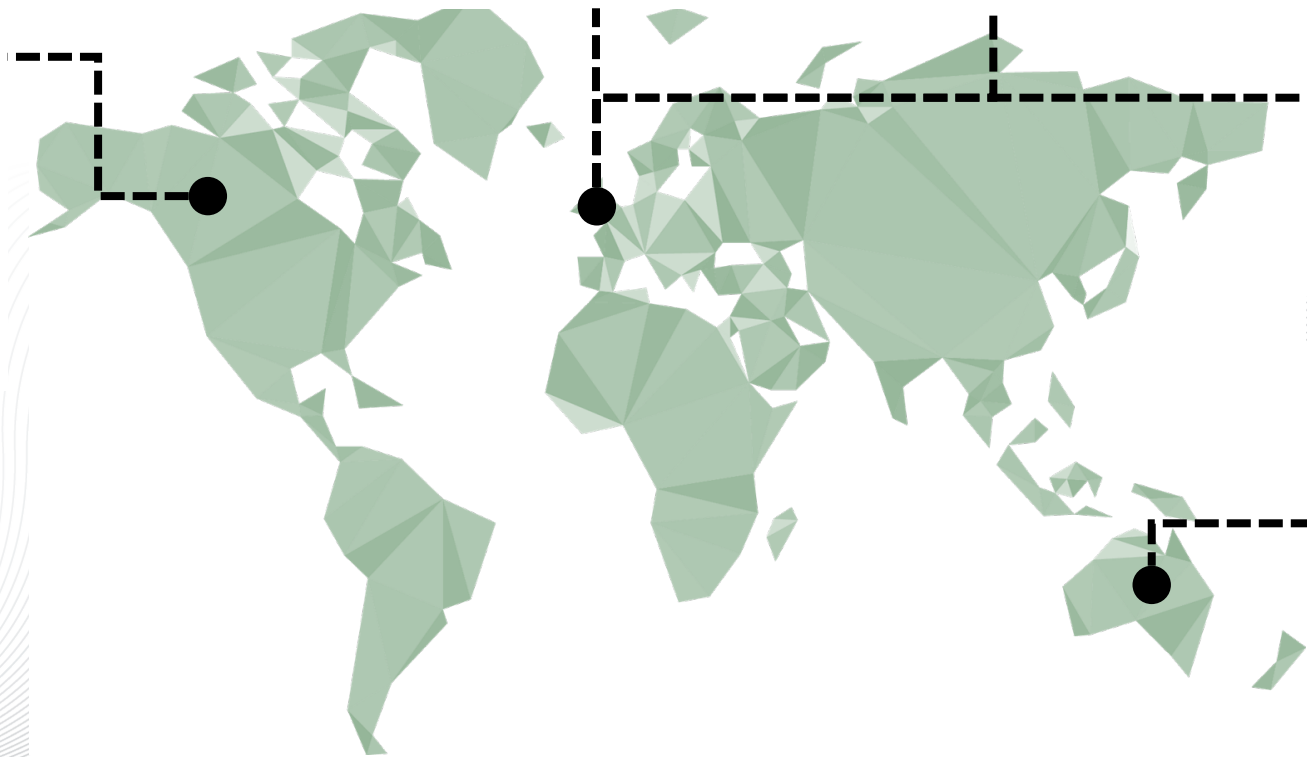
- Large banks exploring the use of **bespoke, dynamic customer messaging as part of the payments journey** – to help educate customers and improve scam action

Efficiency Play: UK

- Large banks investing heavily in exploring **agentic use cases in Fraud & AML operations** for overhead reduction and ROI but reticent of the time it takes for meaningful transformation (i.e., 5-7 years)
- Key use cases for agentic being trialed is for:
 - **Source of wealth identification**
 - **Automated refunds** for simple fraud cases
 - **Investigation support**, determining which sources are useful for the specific investigation

Efficiency & Capability Play: Australia

- Use Cases gaining traction are focused on **increasing automation in regulatory compliance assessments in cyber crime**: digesting annual regulatory updates (e.g., for PCI DSS, SOC, TPRM, Op Resilience, SWIFT), capturing what has changed, what can be leveraged from last year's assessment, what gaps need to be filled, automated evidence logging and follow up, modifying risk based on impact scale.



✓ Efficiency ✓ Experience ✓ Capability

Following the announcement of Budget 2025 in Canada, increasing focus on fraud management will likely drive further investments in the usage of AI for regulatory compliance purposes

IN-SESSION POLL RESULTS!

Where are you **focusing investments** in AI capabilities?

25%

Identity verification &
authentication

29%

Operational uplift of
investigation, disputes and
remediation activities

5%

Attack simulation / red teaming

5%

Other

36%

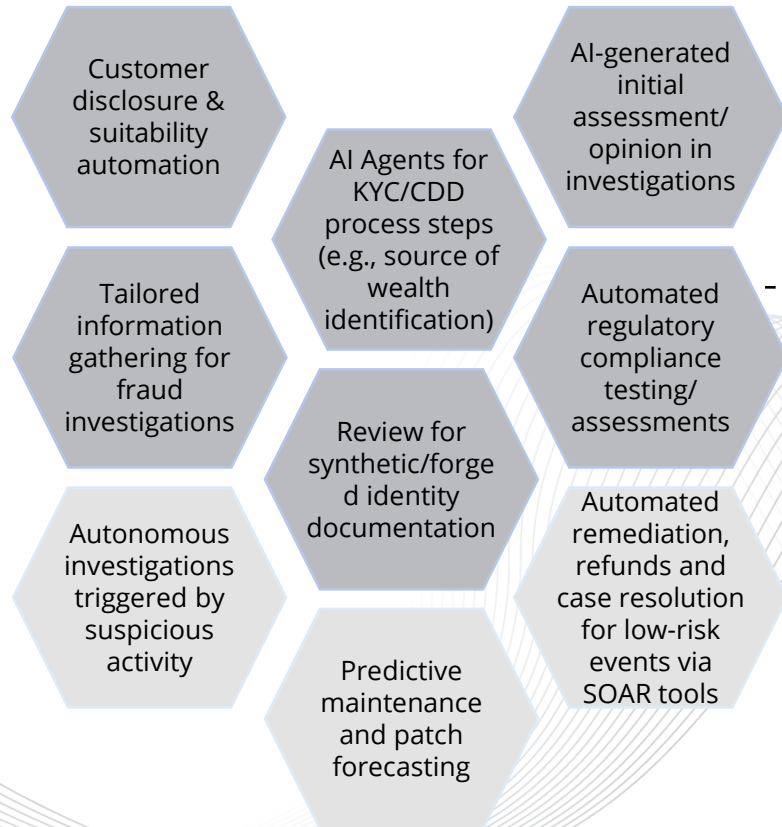
Augmenting existing detection /
monitoring models and
analytics

Organizations leverage use cases to boost efficiency, enhance customer experience and expand capabilities. From a cyber and fraud lens, AI-led use cases primarily focus on efficiency and capability plays

AI Use Case Repository

✓ EFFICIENCY PLAY

Improve operational performance by doing more with less



Threat Detection and Monitoring

Fraud/Cyber compliance & operational efficiency

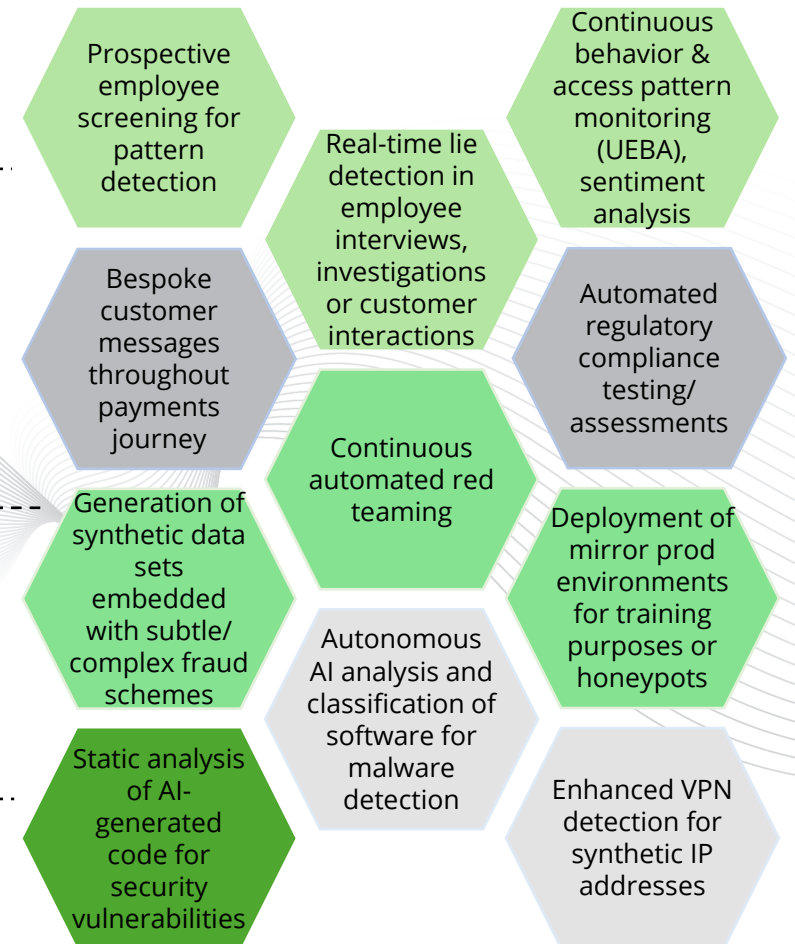
Large-scale synthetic attack simulation capabilities

Adaptive defenses for evolving threats and system vulnerabilities

Security testing capabilities prior to integration

✓ CAPABILITY PLAY

Develop and enhance capabilities that are enabled by AI



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How can Deloitte's Financial Crime Practice help?

Deloitte helps clients design, build, and operate dynamic, business-aligned fraud programs for various lines of businesses. You can be assured of being well equipped to meet the requirements, while continuing to focus on what you do best: managing your organization

Global leadership: Deloitte is a global leader in Fraud Strategy Consulting. We offer differentiated domain leadership and entrenched industry experience

Ecosystems and alliances: Strong alliances with leading technology vendors, industry organizations, and research entities provide leading insights, intelligence, information-sharing, and collaboration

Quantification of fraud risk: Through integrating data and tailored statistical models for risk quantification, we help organizations to augment their experience with technology to develop risk-intelligence responses

Data-driven solutions: Through our deep technical and industry insight with cutting-edge analytics, we help identify the right data-driven fraud solutions to address the most complex issues impacting your business today

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